

2-Day Implementation Plan

Team of 2 · Target: working MVP demo for hackathon judging

Hour 0 — Before you write any code (both, ~30 min)

- Register for Google Earth Engine (Community quota tier, noncommercial, decision-making use case).
- Generate a free Gemini API key via Google AI Studio.
- Lock in the demo storm: pick ONE historical cyclone with clean IBTrACS data and known impact (Amphan 2020 or Remal 2024 recommended). Do not revisit this decision later.
- Create the shared repo, agree on the GeoJSON schema for risk output up front so both tracks can work in parallel without blocking each other.

Day 1

Time block	Person A (Pro) — Data / Backend	Person B (Noob) — Frontend / UI
Morning (~4 hrs)	Pull IBTrACS track for chosen cyclone → clean into lat/lon with time sequence project (Miles/Foots) API. Map the one Pull JS/Python (5	
Afternoon (~4 hrs)	Pull roads/hospitals/power infra from Overpass API for the chosen line. Display on map (test the risk scoring logic hold ese	
Evening (~2 hrs)	Expose a simple REST endpoint that serves the per-timestep GeoJSON. Safety for back Rest API against spam (even phishy oper	

Day 2

Time block	Person A (Pro) — Data / Backend / LLM	Person B (Noob) — Frontend / Polish
Morning (~3 hrs)	Integrate Gemini 3.7 Flash: feed it the top-N highest-risk coastal GeoJSON, combining into the map (JS/Python, replace the doc). Get	
Midday (~2 hrs)	Add the 'advisory dispatch' as a mocked preview panel (e.g. generate SMS/email text) ring in for tools, read will be in reg	
Afternoon (~2 hrs)	Full integration testing with Person B. Fix data-contradictions. Verify with Person A. Dry-run case research. Confir	
Evening (~2 hrs)	Write the 2-minute pitch script, mapped directly to the README. Tailor demo to your languages (python and js are ok). Final revisi	

Cut-list — drop these first if you're behind schedule

- Real storm-surge physics simulation (use the heuristic formula instead, clearly labeled as such).
- Live/real-time weather feeds (historical replay is fine and more reliable for a demo).
- Real SMS/email dispatch integration (a mocked preview panel is enough).
- Multi-cyclone support (one well-executed storm beats three half-working ones).
- Any login/auth system (not needed for a demo).

- The optional RAG-lite protocol-grounding layer — only add this if everything else is done with time to spare.